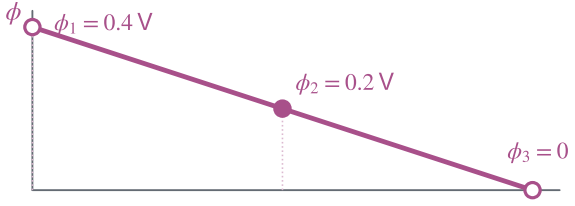
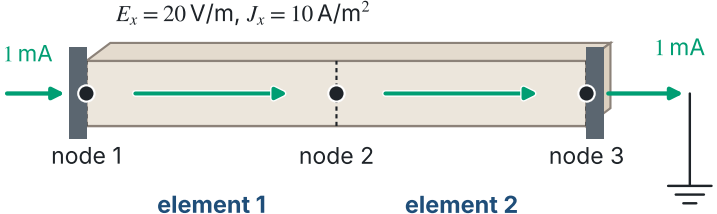
two potentials given, one unknown left





$\mathbf{K}\boldsymbol{\phi} =$ injected node currents

$$5\,\text{mS} \times \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.4 \\ \mathbf{0.2} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{+1} \\ \mathbf{0} \\ \mathbf{-1} \end{bmatrix}$$

V mA

row 2: the equation for ϕ_2 (current balance at node 2)

rows 1, 3: evaluated afterwards,
giving the terminal currents (reactions)

one power, three ways

$$VI = \boldsymbol{\phi}^{\mathrm{T}}\mathbf{K}\boldsymbol{\phi} = \int \sigma A\,(\mathrm{d}\phi/\mathrm{d}x)^2\,\mathrm{d}x = 0.4\,\text{mW}$$